



# Midterm I Exam

**Course:** Electromagnetic Theory  
**Instructor:** Dr. Shafayat Abrar  
**Marks:** 50 points [15% of the total]

**Date:** Feb. 27, 2018  
**Term:** Spring 18  
**Duration:** 2 hours

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## Instructions:

1. Attempt all questions. Formula-sheet is allowed.
2. Remember that talking is not allowed at any time in the exam hall.
3. All sort of nature call may be served only during last fifteen minutes.
4. Stop writing when the teacher says it is the end of the exam.

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1. (05 points) Use Gauss's law to find the electric field inside a uniformly charged solid sphere (charge density  $\rho$ ) of radius  $R$ . Draw the magnitude of  $\mathbf{E}$  versus  $r$ .

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2. (05 points) A simple classical model of an atom consists of a nucleus of a positive charge  $q = N|e|$  surrounded by a spherical electron cloud of the same total negative charge. ( $N$  is the atomic number and  $e$  is the electronic charge.) An external electric field  $\mathbf{E}_o$ , will cause the nucleus to be displaced a distance  $r_o$ , from the center of the electron cloud, thus polarizing the atom. Assuming a uniform charge distribution within the electron cloud of radius  $b$ , find  $r_o$ .

Hint: Use the answer of Q1 to find the force experienced by nucleus at  $r_o$  due to the (spherical) electron cloud. Now exploit the Newton's third law that the external and internal forces must balance each other at the displaced position  $r_o$ .

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3. (05 points) Given  $\mathbf{F} = xy\hat{\mathbf{x}} - 2x\hat{\mathbf{y}}$ , verify Stokes's theorem over a quarter circular disk with a radius 3 in the first quadrant.

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4. (05 points) Given a vector field

$$\mathbf{F} = 3y\hat{\mathbf{x}} + (3x - 2z)\hat{\mathbf{y}} - (2y + z)\hat{\mathbf{z}}$$

Determine the scalar potential function  $V$  whose negative gradient equals  $\mathbf{F}$ .

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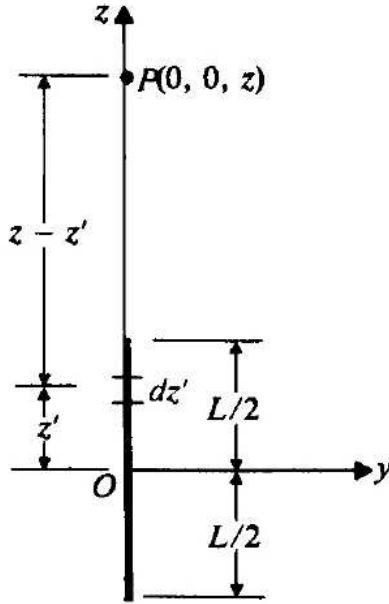


Figure 1: Figure for Q7

5. (05 points) Given the two vectors  $\mathbf{A}$  and  $\mathbf{B}$ , how do you find (a) the component of  $\mathbf{A}$  in the direction of  $\mathbf{B}$ , and (b) the component of  $\mathbf{B}$  in the direction of  $\mathbf{A}$ .

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6. (05 points) Given a function  $\mathbf{E} = y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$ , evaluate the line integral  $\int \mathbf{E} \cdot d\mathbf{l}$  from  $P_1(2, 1, -1)$  to  $P_2(8, 2, -1)$  along the parabola  $x = 2y^2$ .

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7. (10 points) Obtain the expressions for the electric potential and electric field intensity along the axis of a uniform line charge of length  $L$  with density  $\lambda$ .

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8. (05 points) Two positive charges,  $Q_1$  and  $Q_2$ , are located at  $(1, 2, 0)$  and  $(2, 0, 0)$ , respectively. Find the relation between  $Q_1$  and  $Q_2$  such that the total force on the test charge at point  $P(-1, 1, 0)$  will have (a) no  $x$ -component, (b) no  $y$ -component.

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9. (05 points) A line charge of uniform density  $\lambda$ , in free space forms a semicircle of radius  $b$ . Determine the magnitude and direction of the electric field intensity at the center of the semicircle.